

# Exosome-Based Therapeutics: A Natural Solution to Overcoming the Blood-Brain Barrier in Neurodegenerative Diseases

Pharmaceutics 2025

**Abstract record only.** No openly available full-text PDF was found for this paper (it appears to be behind a paywall or not deposited in an open repository). This file contains the bibliographic record and the abstract. Access the publisher version via the DOI above, e.g. through your institution.

**Journal:** Pharmaceutics

**Year:** 2025

**Citation as given:** Pharmaceutics. 2025

**DOI:** 10.1002/mco2.70386 | **Link:** <https://doi.org/10.1002/mco2.70386>

## Abstract

Neurodegenerative diseases present significant therapeutic challenges, primarily due to the restrictive nature of the blood–brain barrier (BBB), which limits drug delivery to the brain. While the BBB is crucial for protecting the brain from harmful substances, it also hinders the effectiveness of treatments for neurodegenerative diseases. Consequently, there is an urgent need for innovative drug delivery systems capable of bypassing the BBB to improve therapeutic outcomes. Exosomes, as endogenous nanoscale carriers, offer substantial promise for brain-targeted drug delivery. Their unique characteristics, including the ability to cross biological barriers, high biocompatibility, intrinsic targeting capacity, natural intracellular transport mechanisms, and robust stability, render them highly promising candidates for drug delivery in the treatment of neurodegenerative disorders. This review delves into various engineering strategies for exosome-mediated targeted drug delivery and provides an in-depth analysis of the structural and functional properties of the BBB under normal and pathological conditions. We emphasize the potential of exosomes as drug delivery vehicles for the central nervous system, particularly in addressing neurodegenerative disorders. Furthermore, we address the key obstacles to the clinical application of exosome-based therapies and propose future research directions aimed at optimizing these methods to develop more effective treatment strategies.